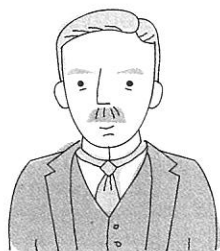


Science as a human endeavour

 Visual/spatial  Verbal/linguistic

Scientists have developed models to represent atoms. Refer to the Science as a Human Endeavour on pages 8–9 of your student book to read more about the development of atomic models.

- 1 **Match** the scientist with the atomic model that made him famous. The first one has been done for you.



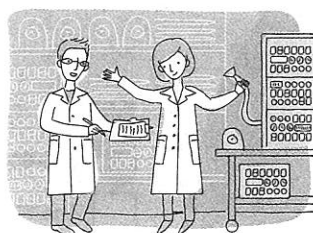
A Ernest Rutherford



B Neils Bohr



C Sir James Chadwick



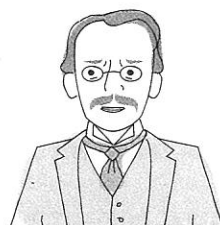
D Modern atomic physicists



E Democritus



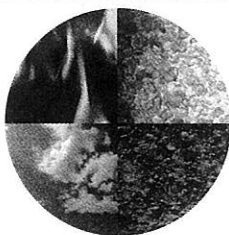
F Ancient Greek philosophers



G Sir Joseph J. Thomson



H Philipp Lenard



Continuum model

Substances are made up of the four elements: air, fire, water and earth. Substances can always be broken down into smaller pieces of the same substance.

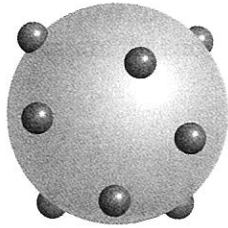
Scientist: Ancient Greek philosophers



Hard sphere model

Substances are made up of hard, ball-like building blocks called atoms that cannot be broken apart.

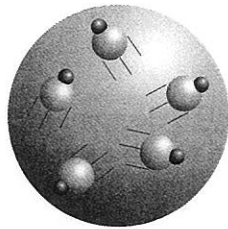
Scientist: _____



Plum pudding model

Atoms are made up of a positively charged ball with negatively charged electrons stuck in it.

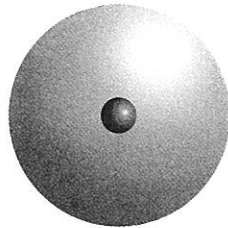
Scientist: _____



Dynamide model

Atoms are made up of moving particles called dynamides. Each dynamide is made up of a positively charged particle and a negatively charged particle.

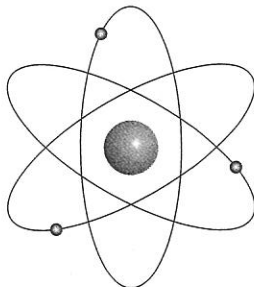
Scientist: _____



Nuclear model

Atoms are made up of a solid, positively charged nucleus surrounded by an electron cloud.

Scientist: _____

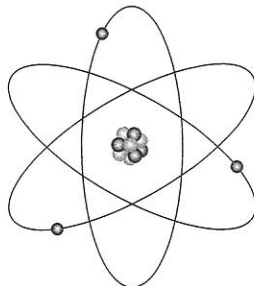


Planetary model

Atoms are made up of a solid, positively charged nucleus surrounded by electrons orbiting the nucleus, like planets orbiting the Sun.

Scientist: _____

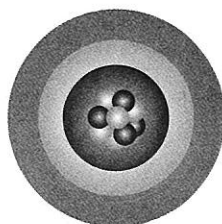
orbit (v) to move around something in a circle or ellipse



Neutron planetary model

Atoms are made up of a positive nucleus that has both protons and neutrons. The nucleus is surrounded by electrons orbiting it, like planets orbiting the Sun.

Scientist: _____



Electron shell model

Atoms are made up of a positive nucleus that has both protons and neutrons. The electrons form shells around the nucleus.

Scientist: _____